

João Pedro Paiva Pires



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I am currently pursuing an MSc in Data Science and Engineering at IST. My background in Physics gave me a strong analytical foundation, but it was through real-world projects that I found my passion, from building deep learning models to predict stock market trends, to co-authoring research on ML-based battery monitoring, currently under review at Nature Machine Intelligence. I am now looking to bring these skills into the financial sector, where I can grow at the intersection of data science and quantitative finance.

Education

University of Aveiro, BSc in Physics Sept 2022 – July 2025

- **Coursework:** Calculus, Linear Algebra, Discrete Mathematics, Numerical Methods, Classical Mechanics, Statistical Physics, Quantum Mechanics, Electrodynamics, Solid State Physics, Atomic and Nuclear Physics.
- **Research Grant Award** – Recognized with a research grant during my final year, declined due to institutional exclusivity.

Instituto Superior Técnico, MSc in Data Science and Engineering Sept 2025 – July 2027

- **Coursework:** Machine Learning, Deep Learning, Computational Statistics, Network Science, Information Systems and Databases.

Publications

"Beyond the Surface: Estimating Internal Battery Pressure and Temperature with Optical Fiber Sensors and Machine Learning", *Nature Machine Intelligence* Under-Review
L. Matuck, M. Nascimento, J. P. Pires, J. F. F. Mendes, M. S. Ferreira

- **Data Engineering:** Built a preprocessing pipeline (Savitzky–Golay denoising + Z-score normalization) across 34 LIB and 27 SIB charge–discharge cycles.
- **Model Optimization:** Tuned chemistry-specific TCN architectures via Bayesian optimization (500 trials), achieving temperature RMSE of 0.075°C (LIB) and pressure RMSE of 0.048 bar.
- **Machine Learning:** Trained TCN virtual sensors to reconstruct internal battery states from non-invasive signals only, reaching R^2 of 0.996 (temperature) and 0.97 (pressure) for LIB, developing research maturity and technical autonomy as an undergraduate collaborating on doctoral-level work.

Projects

Deep Learning Architectures for Predicting PSI20 Trends github.com/joaoopiress

- Developed and compared LSTM, ALSTM, and Transformer models to classify Portuguese stock index movements, achieving a Macro F1-Score of 62.5%.
- **Data Engineering:** Engineered a pipeline using Wavelet Transform denoising and Z-score normalization on a 10-year dataset to enhance signal-to-noise ratio.
- **Model Optimization & Award:** Used Bayesian Search to refine model parameters, achieving **18/20** and **Best Project**, an entirely self-driven project outside the Physics curriculum that built resilience and self-learning skills.

Personal Finance Tracker github.com/joaoopiress

- Built a personal expense tracking tool that categorizes monthly spending across 10 categories and forecasts next-month expenses using a predictive ML model. Used NLP for real-time category recognition.

Extracurricular Activities

Mentor in the UA Mentorship Program, University of Aveiro – Aveiro, PT Sept 2024 – June 2025

- Mentored 15 first-year students throughout the full academic year, guiding them on course selection, study strategies, and career path orientation through regular sessions.
- Developed leadership and communication skills by fostering a supportive environment that helped students navigate both academic and personal challenges of university life.

Technical Skills

Programming Languages & Databases: Python, Matlab, R, SQL (PostgreSQL).

ML Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face.

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly.

Other Tools: Git, LaTeX, Jupyter, Google Colab, Office 365.